

NEX Clinical Intelligence System

AI-Powered Patient Intake Automation Platform
A Meridux Product

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10

Automated Stages

0

Manual Steps Required

<30s

Response Time

100%

Data Ownership

OVERVIEW

NEX Clinical Intelligence System is a fully automated patient intake platform built on n8n. It eliminates manual processing by handling every step of the intake process — from form submission to confirmation email — without any human intervention.

Designed for healthcare providers spending hours every week on manual data entry, duplicate checking, and patient follow-ups, NEX delivers a complete automated operations layer that runs 24/7 and logs every interaction to a Google Sheets dashboard the clinic owns outright.

THE PROBLEM THIS SOLVES

PROBLEM	SOLUTION
Manual data entry for every patient intake form	Fully automated — data captured and logged instantly
Duplicate patient records created by simultaneous submissions	Race condition lock prevents duplicate processing at source
No-shows due to slow or missed confirmation emails	Automated HTML confirmation email sent within 30 seconds
Staff spending hours building weekly patient reports	Scheduled daily summary email auto-generated at set time
Patient data locked inside third-party SaaS platforms	All data stored in clinic-owned Google Sheets — no lock-in

How NEX Processes Every Patient Intake

01	Intake Trigger Tally form submission fires a webhook to n8n, capturing all patient data fields instantly.
02	Race Condition Lock JavaScript code node builds a unique submission key and checks an active locks sheet — prevents duplicate processing on simultaneous submissions.
03	Duplicate Detection Cross-references incoming data against existing Google Sheets records to identify returning patients vs new entries.
04	Patient History Lookup Queries patient history log to retrieve prior visit data, flagging returning patients for conditional routing.
05	Returning Patient Check Applies conditional logic — routes returning patients through a fast-track path, new patients through full onboarding flow.
06	Main Data Processing Normalises and structures all intake data using JavaScript code nodes — timestamps, unique ID assignment, and field validation.
07	Eligibility Gate Checks patient eligibility criteria. Approved submissions proceed to confirmation; flagged submissions route to manual review queue.
08	Booking Confirmation Sends an HTML-formatted confirmation email via Gmail within 30 seconds — dynamically populated with patient-specific details.
09	Encounter Logging Logs the full processed intake record to Google Sheets with timestamp, unique ID, status, and processing metadata.
10	Daily Summary Report Scheduled workflow fires at end of day — aggregates all encounters and emails an executive summary to clinic management.

TECHNOLOGY STACK

n8n Workflow Automation Engine Self-hosted	Tally Forms Patient Intake Forms Webhook trigger	Google Sheets Data Storage & Logging Clinic-owned
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Pinecone Vector Database AI lookups	Gmail API Automated Email Delivery HTML templates	JavaScript Custom Logic Nodes Code nodes
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What Makes NEX Different

Race Condition Lock

A JavaScript-built unique key system prevents duplicate records even when multiple patients submit simultaneously — without any external database or locking service. Built entirely within n8n using code nodes and a dedicated locks sheet in Google Sheets.

Zero SaaS Dependency

All patient data is stored in a Google Sheets file the clinic owns outright. No monthly SaaS subscription, no vendor lock-in, no third-party data access. The clinic controls their data permanently.

Dynamic HTML Email Templates

Every confirmation email is uniquely generated per patient — appointment type, provider, date, time, and urgency level are dynamically populated from intake data. Not a generic template.

Conditional Routing Intelligence

Returning patients are automatically routed through a fast-track path based on history lookup results. New patients flow through full onboarding. Ineligible submissions are flagged for manual review without blocking the main flow.

Executive Daily Summary

A scheduled workflow fires at end of day, aggregates all encounters, and emails a management summary automatically. Zero manual reporting effort required.

Fully Adaptable Architecture

While built for healthcare intake, the underlying architecture — webhook trigger, race lock, duplicate detection, conditional routing, automated response, encounter logging, scheduled reporting — is directly applicable to any service business handling inbound enquiries.

RESULTS & IMPACT

10–20+

Staff hours recovered per week

<30s

Patient confirmation delivery time

0

Duplicate records from race conditions

24/7

System operates without human intervention

Want This Built for Your Business?

This system is fully adaptable for any service business handling inbound enquiries — healthcare, professional services, real estate, finance, and beyond.

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